Do Now · retrieval

Lesson 087 · Algorithm Design and Testing Challenge

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is an invariant?

Q2

Why is sorting unnecessary for just finding one maximum?

Q3

What is nullable int?

Q4

Binary search for 8 in [2,4,6,8,10]. Give midpoint indexes and result.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — Specify output when inputs are empty, and preserve correctness on all-negative arrays.

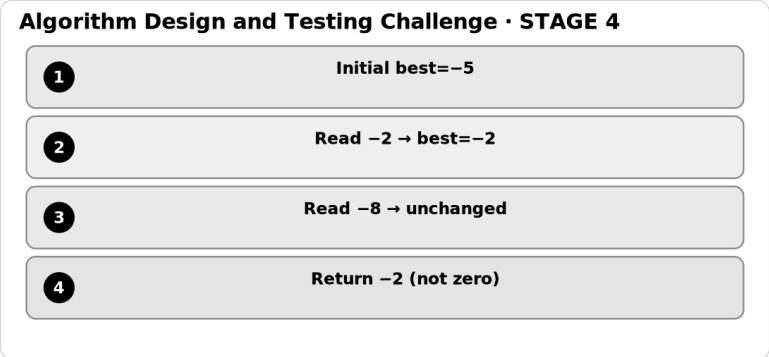
Core idea 2

Explain — Maintain invariant: best is the largest value among those inspected so far.

Core idea 3

Evaluate — Compare against alternative sorting-based solution: an O(n) scan avoids O(n log n) sort when only the maximum is needed.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

maximum — Largest value in set.

invariant — Maintained fact during loop.

nullable — Type allowing absence/null.

complexity — Resource growth as input increases.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Trace Maximum([7,3,10,4]).

Follow the worked method

1. Model response — The invariant tracks largest seen, not last examined.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Model response
Starting best at0 would be incorrect for all-negative arrays.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace Maximum([-5,-2,-8]) showing all changes and final value. [6]

Question 2

Write algorithm for maximum of array, including empty input case. [7]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare O(n) maximum scan with sorting first and describe useful test coverage. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Initialising maximum to zero works for every integer array.

Use key vocabulary from memory

1. Explain maximum in your own words.

2. Explain invariant in your own words.

3. Explain nullable in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.